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Inventor(s)

Biedermann; Lutz et al.

Receiving part for receiving a rod for coupling the rod to a bone anchoring element

Abstract

A receiving part for receiving a rod for coupling the rod to a bone anchoring element includes a receiving part body including a first end and a second end, and having a substantially U-shaped recess at the first end forming a channel for receiving the rod, and an accommodation space for accommodating a head of the bone anchoring element, the accommodation space having an opening at the second end for introducing the head; and a pressure element arranged at least partially in the accommodation space, the pressure element including a first section having a second recess for receiving the rod, and a second section having a flexible portion to clamp the head, the first section and the second section being fixed relative to each other, wherein said pressure element is insertable from the opening.

Inventors: Biedermann; Lutz (VS-Villingen, DE), Matthis; Wilfried (Weisweil, DE)

Applicant: BIEDERMANN TECHNOLOGIES GMBH & CO. KG (Donaueschingen, DE)

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Assignee: BIEDERMANN TECHNOLOGIES GMBH & CO. KG (Donaueschingen, DE)

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Primary Examiner: Robert; Eduardo C

Assistant Examiner: Negrelli-rodriguez; Christina

Attorney, Agent or Firm: Womble Bond Dickinson (US) LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation of U.S. patent application Ser. No. 17/493,042, filed Oct. 4, 2021, which is a continuation of U.S. patent application Ser. No. 16/538,163, filed Aug. 12, 2019, now U.S. Pat. No. 11,160,582, which is a continuation of U.S. patent application Ser. No. 15/872,428, filed Jan. 16, 2018, now U.S. Pat. No. 10,426,522, which is a continuation of U.S. patent application Ser. No. 14/878,809, filed Oct. 8, 2015, now U.S. Pat. No. 9,895,172, which is a continuation of U.S. patent application Ser. No. 12/855,395, filed Aug. 12, 2010, now U.S. Pat. No. 9,283,000, which claims the benefit of U.S. Provisional Patent Application Ser. No. 61/233,406, filed Aug. 12, 2009, the contents of which are hereby incorporated by reference in their entirety, and claims priority to European Patent Application EP 09 167 751.8, filed Aug. 12, 2009, the contents of which are hereby incorporated by reference in their entirety.

BACKGROUND

Field

(1) The invention relates to a receiving part for receiving a rod for coupling the rod to a bone anchoring element. The receiving part includes a receiving part body with a channel for receiving a rod and an accommodation space for accommodating a head of a bone anchoring element, and a pressure element for clamping the head. The accommodation space has an opening at a bottom side for inserting the pressure element and for inserting the head.

Description of Related Art

(2) Various designs of polyaxial bone screws are known wherein a head is clamped from the side to lock a rotational position of the bone screw.

(3) U.S. Pat. No. 5,672,176 describes a bone screw with a receiving part with a conically shaped seat and a conically shaped pressure element which exerts a pressure onto the head from above and from the side.

(4) U.S. Pat. No. 5,669,911 describes a polyaxial orthopaedic device for use with a rod implant apparatus. The device includes a screw having a curvate head, a locking collar disposed therearound, and a receiving member having a linearly tapered socket in which the screw and the collar are nested. The collar is introduced from the top of the receiving member. The head of the screw can be inserted from the bottom.

(5) U.S. Pat. No. 6,063,090 relates to a device used to connect a longitudinal support to a pedicle screw by an accommodating head having a channel to accommodate the longitudinal support. The pedicle screw and the accommodating head are connected via a conical collate chuck in the accommodating head and by a spherical head on the pedicle screw. The device allows engagement of the pedicle screw in the accommodating head after the pedicle screw has been inserted into the bone.

SUMMARY

(6) Embodiments of the invention provide an improved receiving part for receiving a rod for coupling the rod to a bone anchoring element, and a bone anchoring device with such a receiving part, where the embodiments have fewer parts, a low profile, and provide for improved handling during surgery.

(7) Embodiments of the invention provide a receiving part including a receiving part body including a first end and a second end, and having a substantially U-shaped recess at the first end forming a channel for receiving a rod, and an accommodation space for accommodating a head of a bone anchoring element, the accommodation space having an opening at the second end for introducing the head; and a pressure element arranged at least partially in the accommodation space, the pressure element including a first section having a second recess for receiving the rod, and a second section having a flexible portion to clamp the head, the first section and the second section being fixed relative to each other, wherein said pressure element is insertable from the opening, a bone anchoring device including such a receiving part, and a method of using such a receiving part.

(8) For the receiving part according to an embodiment of the invention, the pressure element has a slim design which allows it to be introduced from the bottom into the receiving part body. An internal end stop within the receiving part body may form an abutment for the pressure element to be positioned in an insertion position for the head. Therefore, there may be no additional components for holding the pressure element in the insertion position.

(9) In some embodiments, the receiving part has a low profile and improved or maximum stiffness, since a wall thickness of the receiving part body can be increased due to the slim design of the pressure element.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further features and advantages of the invention will become apparent from the description of embodiments using the accompanying drawings. In the drawings:

(2) FIG. 1 shows a perspective exploded view of a first embodiment of the bone anchoring device.

(3) FIG. 2 shows a perspective view of the bone anchoring device of FIG. 1 in an assembled state.

(4) FIG. 3 shows a cross-sectional view of the bone anchoring device of FIG. 1, the section being taken perpendicular to a rod axis.

(5) FIG. 4 shows an enlarged perspective view of a pressure element according to an embodiment of the bone anchoring device.

(6) FIG. 5 shows a cross-sectional view of the pressure element according to FIG. 4.

(7) FIG. 6 shows a side view of the pressure element of FIG. 4.

(8) FIG. 7 shows a bottom view of the pressure element of FIG. 4.

(9) FIGS. 8a to 8d show schematic views of design modifications of a pressure element and a receiving part body in a locking position according to embodiments of the bone anchoring device.

(10) FIGS. 9a and 9b show schematic views of a further design modification of the pressure element and the receiving portion in a pre-locking position and a locking position, respectively, according to an embodiment of the bone anchoring device.

DETAILED DESCRIPTION

(11) As shown in FIGS. 1 and 2, a bone anchoring device according to a first embodiment includes a bone anchoring element 1 in the form of a bone screw having a threaded shaft 2 and a head 3, which in this embodiment is a spherical segment-shaped head. The head 3 has a recess 4 for engagement with a screwing-in tool. The bone anchoring device further includes a receiving part body 5 for receiving a rod 6 to connect the rod to the bone anchoring element 1. Further, a closure element 7, in the form of an inner screw or set screw in some embodiments, is provided for

securing the rod **6** in the receiving part body **5**. In addition, the bone anchoring device includes a pressure element **8** for locking the head **3** in the receiving part body **5**.

(12) The receiving part body **5** is now explained with reference to FIGS. **1** to **3**. The receiving part body **5** includes a first end **5a** and a second end **5b**, and an axis of symmetry **M** passes through the first and the second end. A bore **9**, which is coaxial with the axis of symmetry **M**, extends from the first end **5a** to the second end **5b**. The bore **9** may have a smallest diameter at, about, or adjacent to its middle section **91** in a center region of the receiving part body **5**. In a first region adjacent to the first end **5a**, the receiving part body **5** has a U-shaped recess **10**, which may be symmetric with respect to the symmetry axis **M**. The recess **10** has a bottom which may be directed towards the second end **5b** and provides two free lateral legs **11a**, **11b** extending towards the first end **5a**. In the region of the legs **11a**, **11b** an internal thread **12** may be provided which may cooperate with the closure element **7**, which in this embodiment is a set screw. A channel formed by the U-shaped recess **10** is sized so as to receive the rod **6** therein, the rod for connecting a plurality of bone anchoring devices.

(13) At a second region near the second end **5b**, the receiving part body **5** has a narrowing portion **13**, which narrows in a direction towards the second end **5b**. The narrowing portion **13** provides a seat for the screw head and the pressure element. The narrowing portion **13** is located at a distance from the second end **5b** to, for example, enable the screw shaft **2** to be pivoted in a certain pivot angle range. An opening **14** is provided at the second end **5b**, the diameter of which may be the same as or larger than the diameter of the narrowing portion **13** in some embodiments. Between the narrowing portion **13** and the middle portion **91** of the bore **9**, a portion with an inner diameter larger than the diameter of the narrowing portion **13** is provided, which forms a space **15** for allowing the pressure element to expand, as described later. At its side opposite to the narrowing portion **13**, the space **15** connects to the middle section **91** of the bore **9** having the smaller diameter, thereby providing a circular shoulder **16**.

(14) The middle section **91** of the bore **9** includes a circular projection **17** at its side opposite to the space **15**, which can also act as a stop, as described below in more detail.

(15) As can be seen in FIGS. **1** and **3** to **7**, a pressure element **8** according to an embodiment of the invention includes a first section **81** which is substantially cylindrical and has an outer diameter which may be slightly smaller than the inner diameter of the middle section **91** of the bore **9**, so that the pressure element **8** is movable in the bore **9**. It further has a second section **82**, which has a hollow interior **83** which is substantially spherically-shaped and is sized to clamp the spherical head **3** therein. The outer diameter of the first section **81** may be, as can be seen in FIG. **3**, smaller than an outer diameter of the head **3** and a largest outer diameter of the second section **82** is greater than the outer diameter of the cylindrical section **81**. A free end of the second section **82** provides an opening **84** for introduction of the head **3**. Further, the second section **82** may have a plurality of slits **85** extending from the edge of the opening **84** through the second section **82** to define or form slightly resilient legs **90**. The number and the dimension of the slits **85** may be such that the wall of the second portion is flexible enough to snap onto the head **3** when the head **3** is being inserted. The slits **85** can extend into the first cylindrical section **81** as shown in the figures to enhance flexibility. The outer wall surface of the second section **82** may include a first portion **86** which may be spherically-shaped, and a second portion **87** adjacent to the opening **84** which may be tapered or curved or otherwise narrowing towards the opening **84**. The portion **87** cooperates with the narrowing portion **13** of the receiving part body when the head is locked in the receiving part body.

(16) Further, the pressure element has a coaxial bore **88** for providing access to the screw head by a tool. The first section **81** may have at its free end a cylindrically-shaped or cylinder segment-shaped recess **89** for receiving the rod **6** therein.

(17) The dimensions of the pressure element **8** are such that the pressure element **8** can be inserted through the opening **14** at the second end **5b** of the receiving part body **5**, with the cylindrical first section **81** inserted first. When the second section **82** passes through the opening **14**, it may be

compressed due to the flexibility of the second section **82**, or the legs **90** thereof, which allows the pressure element **8** to be fully introduced into the receiving part body **5**.

(18) The pressure element **8** can be pushed into the receiving part body **5** until an upper edge of the first section **81** abuts against the stop **17** provided by the annular projection in the receiving part body **5**. When the pressure element is in this position, the flexible second section **82** is located in the portion between the middle section **91** of bore **9** and the narrowing portion **13** of the receiving part body **5**. In this position, a free space in the space **15** between the outer wall of the second section **82** of the pressure element and the inner wall of the receiving part body **5** provides room for the flexible second section **82** to expand when the head **3** is inserted.

(19) A depth of the cylindrical recess **89** of the pressure element is smaller than a radius of the rod **6**, so that the pressure element can be pressed down by means of screwing-in the inner screw **7**, which presses onto the rod **6**, which in turn presses onto the pressure element **8**. The pressure element **8** is oriented in the receiving part body **5** such that its cylindrical recess **89** is aligned with the U-shaped recess of the receiving part body **5**.

(20) The material of which the components of the bone anchoring device are made is preferably a body compatible metal, such as stainless steel or titanium, or a body compatible alloy such as a nickel titanium alloy, in particular Nitinol. However, body compatible plastic materials, such as medical-grade polyether ether ketone (PEEK), can also be used.

(21) In use, first, the pressure element **8** is introduced via the bottom opening **14** into the receiving part body **5** until it abuts against stop **17**. This can be done either by the surgeon, or before so that the receiving part body is preassembled with the pressure element **8**. Thereafter, the head **3** is introduced through the bottom opening and pushes the pressure element **8** upwards against stop **17**. This is the insertion position, which allows the screw head **3** to be introduced through the opening **84** into the pressure element **8**, thereby widening the hollow interior **83**, or the legs **90** respectively, until the pressure element **8** snaps onto the head **3**. This can be done either before screwing the threaded shaft **2** into the bone, or after having screwed in the threaded shaft **2** into the bone, to mount the receiving part body **5**, with the pressure element inside, onto the head **3**. When the head **3** is inserted, it can still pivot within the pressure element **8** if the flexible section **82** is not compressed. A plurality of bone anchoring devices may be implanted into the bone(s), for example, in the pedicles of adjacent vertebrae, and the rod **6** may be inserted into the bone anchoring devices, respectively. Angular positions of the receiving part bodies may then be adjusted. Finally, the set screw **7** for each bone anchoring device may be tightened, thereby pressing down the rod **6** onto the respective pressure elements, which are pressed down until the second portion **87** of the outer wall of the second section **82** of each pressure element engages with the narrowing portion **13** of the respective receiving part bodies. In this condition, the head of each bone anchoring device is clamped in an interior of the pressure element, so that it is locked in a desired angular position.

(22) FIGS. **8a** to **8d** show various designs of the narrowing portion **13** of the receiving part body **5** and the portion **87** of the pressure element **8** which engages the narrowing portion **13** according to embodiments of the invention. FIG. **8a** shows the two portions tapering lineally at substantially a same angle. This provides a substantially even pressure distribution between the second portion **87** of the pressure element **8** and the narrowing portion **13** of the receiving part body **5**. FIG. **8b** and FIG. **8c** show two different designs where the respective portions are tapered at different angles. FIG. **8b** shows a main contact area at a bottom of the narrowing portion **13**, and FIG. **8c** shows a main contact area at a top of the narrowing portion **13**. FIG. **8d** shows a tapering second portion **87** of the pressure element **8** and a rounded portion **13** of the receiving part body, wherein a curvature of the rounded portion is directed towards a center of the receiving part body **5**. With such a configuration, a contact area positioned at the rounded portion can be achieved.

(23) In FIGS. **9a** and **9b** a narrowing portion **13'** of a modified example of a receiving part body **5** has a double spherical radius formed by two curved portions **13a**, **13b** on top of (i.e., adjacent to) each other, with a groove **13c** therebetween. A curvature of the curved portions are directed towards

the central axis M. Correspondingly, a modified pressure element **8'** has at its lower end two invertedly curved portions **8a**, **8b** which correspond to the curved portions **13a**, **13b**, with a crest **8c** therebetween, and an outer crest **8d** at an outer yielding edge of the pressure element **8'**.

(24) As shown in FIG. **9a**, when the pressure element **8'** moves downward, its lower most edge **8d** engages in the groove **13c**. In this position there is a frictional clamping of the head **3**, which still allows the head **3** to be pivoted upon exertion of a force, where this force is greater than a force needed to pivot the head **3** when the head **3** is introduced in an insertion position. This may be characterized as a pre-locking condition. As shown in FIG. **9b**, by further pressing the pressure element **8'** downward, the curvatures of the pressure element **8'** engage with the corresponding curvatures of the narrowing portion **13'** to finally lock the head **3**.

(25) Further modifications of the bone anchoring device are possible. For example, in one embodiment the pressure element **8** can have a recess corresponding to the cylindrical recess **89** described above, which may be U-shaped and provides legs extending above the rod. A dual part closure element can then be used to separately clamp the head and the rod. A device for preventing the pressure element from rotating can be provided (not shown). Such a device can be realised, for example, by crimp bores or by a pin extending from the wall into a recess (not shown) of the pressure element.

(26) Further, in some embodiments, the abutment or stop **17**, on which a pressure element may abut after it is inserted through the bottom opening and pushed further inwards, can be provided at other locations in the receiving part, for example, at the circular shoulder **16**, which may then interact with a corresponding projecting part of the pressure element to provide an abutment for the pressure element.

(27) While the present invention has been described in connection with certain exemplary embodiments, it is to be understood that the invention is not limited to the disclosed embodiments, but is instead intended to cover various modifications and equivalent arrangements included within the spirit and scope of the appended claims, and equivalents thereof.

Claims

1. A bone anchoring device comprising: a bone anchoring element comprising a shaft and a head; a receiving part for receiving a rod for coupling the rod to the bone anchoring element, the receiving part comprising: a receiving part body having a first end, a second end below the first end, a channel at the first end for receiving the rod, an engagement surface for engagement with a separate closure element for locking the rod in the channel, an accommodation space for accommodating the head of the bone anchoring element, and an opening at the second end sized for inserting the head therethrough into the accommodation space; and a pressure element insertable upwards through the opening of the receiving part body and configured to be positioned at least partially in the accommodation space, the pressure element comprising a rod contacting surface configured to extend into the channel of the receiving part body to contact the rod; wherein when the pressure element is in the receiving part body at an insertion position of the receiving part, the head is insertable upwardly past the opening of the receiving part body to a seat that directly engages the head to restrict the head from falling back out of the receiving part body through the opening; and wherein the receiving part is adjustable from the insertion position to a pre-locking position where the head is prevented from being removed from the receiving part body through the opening and where the receiving part is restricted from being adjusted back towards the insertion position.
2. The bone anchoring device of claim 1, wherein the pressure element further comprises a head contacting surface for contacting and exerting pressure on the head.
3. The bone anchoring device of claim 2, wherein the head contacting surface of the pressure element forms the seat for the head.

4. The bone anchoring device of claim 3, wherein the pressure element is formed as a monolithic part.
5. The bone anchoring device of claim 1, wherein the seat is radially expandable for passing the head therethrough.
6. The bone anchoring device of claim 5, wherein when the receiving part is at the pre-locking position, the seat is restricted from expanding to prevent the head from being removed from the receiving part body.
7. The bone anchoring device of claim 1, wherein when the receiving part is at the pre-locking position, an abutment is configured to hold the pressure element against movement back towards the insertion position.
8. The bone anchoring device of claim 7, wherein the abutment is formed on the receiving part body.
9. The bone anchoring device of claim 1, wherein the rod contacting surface of the pressure element forms a U-shaped recess with an axial depth that is less than half of a greatest width of the U-shaped recess.
10. The bone anchoring device of claim 1, wherein when the receiving part is at the insertion position and the head is held in the seat, the head remains separable from the seat and removable from the receiving part body through the opening.
11. The bone anchoring device of claim 1, wherein the receiving part body defines a longitudinal axis that extends between the first and second ends, and wherein the pressure element and the head are positionable in the receiving part body such that part of the pressure element is at a same axial height as a greatest diameter of the head measured in a direction perpendicular to the longitudinal axis.
12. A bone anchoring device comprising: a bone anchoring element comprising a shaft and a head; a receiving part for receiving a rod for coupling the rod to the bone anchoring element, the receiving part comprising: a receiving part body having a first end, a second end below the first end, a longitudinal axis extending between the first and second ends, a channel at the first end for receiving the rod, an engagement surface for engagement with a separate closure element for locking the rod in the channel, an accommodation space for accommodating the head of the bone anchoring element, and an opening at the second end sized for inserting the head therethrough into the accommodation space; and a pressure element insertable upwards through the opening of the receiving part body and configured to be positioned at least partially in the accommodation space, the pressure element comprising a rod contacting surface configured to extend into the channel of the receiving part body to contact the rod; wherein when the pressure element is in the receiving part body at an insertion position of the receiving part, the head is insertable through the opening into the receiving part body to a position where part of the pressure element is at a same axial height as a greatest diameter of the head measured in a direction perpendicular to the longitudinal axis; and wherein the receiving part is adjustable from the insertion position to a pre-locking position where the opening is at least partially obstructed to prevent the head from being removed from the receiving part body through the opening and where the receiving part is restricted from being adjusted back towards the insertion position.
13. The bone anchoring device of claim 12, wherein when the receiving part is at the insertion position, the head is insertable upwardly past the opening of the receiving part body to a seat that directly engages the head to restrict the head from falling back out of the receiving part body through the opening.
14. The bone anchoring device of claim 13, wherein the seat is radially expandable for passing the head therethrough.
15. The bone anchoring device of claim 13, wherein when the receiving part is at the pre-locking position, the seat is configured to at least partially obstruct the opening of the receiving part body to prevent the head from being removed from the receiving part body.

16. The bone anchoring device of claim 13, wherein the pressure element forms the seat for the head.

17. The bone anchoring device of claim 12, wherein the rod contacting surface of the pressure element forms a U-shaped recess with an axial depth that is less than half of a greatest width of the U-shaped recess.

18. The bone anchoring device of claim 12, wherein when the receiving part is at the insertion position and the head is in the receiving part body, the head remains removable from the receiving part body through the opening.

19. A bone anchoring device comprising: a bone anchoring element comprising a shaft and a head; a receiving part for receiving a rod for coupling the rod to the bone anchoring element, the receiving part comprising: a receiving part body having a first end, a second end below the first end, a channel at the first end for receiving the rod, an engagement surface for engagement with a separate closure element for locking the rod in the channel, an accommodation space for accommodating the head of the bone anchoring element, and an opening at the second end sized for inserting the head therethrough into the accommodation space; and a rod receiving portion separable from and positionable in the receiving part body, the rod receiving portion comprising a rod contacting surface configured to extend into the channel of the receiving part body to contact the rod; and an expandable seat portion separable from and positionable in the accommodation space, the seat portion comprising an upwardly facing surface configured to directly engage and exert an upward pressure directed towards the first end of the receiving part body on the head, wherein an outer width of the seat portion in an unbiased state is greater than a maximum outer width of the rod receiving portion; wherein when the rod receiving portion and the seat portion are in the receiving part body at an insertion position of the receiving part, the head is insertable through the opening of the receiving part body past the seat portion and into the accommodation space; and wherein the receiving part is adjustable from the insertion position to a pre-locking position where expansion of the seat portion is restricted to prevent the head from being removed from the receiving part body through the opening and where the receiving part is restricted from being adjusted back towards the insertion position.

20. The bone anchoring device of claim 19, wherein the maximum outer width of the rod receiving portion is smaller than a diameter of the opening of the receiving part body, and wherein the outer width of the seat portion in the unbiased state is larger than the diameter of the opening of the receiving part body.

21. The bone anchoring device of claim 19, wherein the rod receiving portion and the seat portion are both formed as part of a monolithic pressure element.

22. The bone anchoring device of claim 19, wherein the rod contacting surface of the rod receiving portion forms a U-shaped recess with an axial depth that is less than half of a greatest width of the U-shaped recess.
